

A fair binned baseline halves the peak-level advantage in tandem mass spectrometry

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Abstract

Machine-learning methods for tandem mass spectrometry (MS/MS) have moved from fixed-width binned intensity vectors to models that read peaks at full precision, and the resulting gains are commonly attributed to the spectral representation. We decompose one such gain and find that most of it is neither the representation nor anything about the model. On an openly licensed corpus of 414,992 spectra from GNPS, MassBank and MoNA spanning 66 instrument types, over 10,058 held-out molecules and three seeds, a binned baseline trails a full-precision peak-set model by +0.0579 macro AUPRC. Half of that gap closes when the baseline is given the acquisition covariates — collision energy, adduct, polarity, instrument — which the peak-set arms received all along and the binned arm did not. Against a baseline that has them, the remaining +0.0291 splits evenly: +0.0149 (95% CI +0.0080 to +0.0221) for the encoder and +0.0142 (+0.0097 to +0.0187) for the mass axis, a difference that reverses across seeds. A rounding control that snaps peaks to their bin centres, and an attention-free arm over the same tokens, isolate these terms; within the encoder, the switch to per-peak tokens accounts for all of the effect and self-attention for none (−0.0023, interval straddling zero). Hierarchical aggregation over a molecule’s spectra (+0.0179) is larger than either. Performance measured against a binned baseline says less about spectral representation than about how the baseline was built. Code: <https://github.com/xulinpan/cheminformatics>.

Scientific contribution

We introduce controls that decompose the advantage peak-level models hold over binned baselines in MS/MS substructure prediction: a rounding control separating the spectral representation from the encoder, and an attention-free arm separating the peak-token interface from attention. Applying them exposes a larger confound than either measures — the binned baseline could not see the acquisition metadata, and supplying it halves the reported gap. Against a fair baseline the encoder and the mass axis are indistinguishable, attention contributes nothing, and aggregation exceeds both.

Keywords

Tandem mass spectrometry, Structure annotation, Spectral representation, Mass defect, Substructure prediction, Ablation study

Introduction

Assigning a chemical structure to an MS/MS spectrum is the rate-limiting step in untargeted metabolomics. Where the molecule appears in a reference library the problem reduces to spectral

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matching, but most spectra acquired from biological extracts do not [1]. For the remainder, computational annotation relates the spectrum to candidate structures: by in-silico fragmentation and scoring, as in MetFrag [2]; by predicting a molecular fingerprint and searching a structure database, as in CSI:FingerID [3] and MIST [4]; or by learning a spectrum representation whose geometry reflects chemical similarity, as in MS2DeepScore [5] and DreaMS [6].

For most of the past decade these systems took the same input object. The spectrum, physically a sparse list of $(m/z, \text{intensity})$ pairs, was discretised onto a fixed grid of bins and handed to the model as a dense vector. The convention came from spectral library search, where a bin comfortably wider than the instrument’s mass error makes cosine matching robust to calibration drift between laboratories. A 0.5 Da bin spans roughly 1,700 ppm at m/z 300 while a modern time-of-flight instrument delivers about 5 ppm, so everything the instrument resolves inside that window is discarded — including the mass defect, the difference between a fragment’s exact and nominal mass, which distinguishes C_5H_7^+ (67.0542) from $\text{C}_4\text{H}_3\text{O}^+$ (67.0178) and therefore carries elemental composition.

The field has since moved away from naive binning. CSI:FingerID scores against fragmentation trees over exact masses, MIST tokenises peaks by chemical formula, DreaMS operates on peak lists, and TransExION aligns fragments by nominal mass with the defect embedded as a separate channel [14]; the direction is surveyed in [11], and the convention of discretising a spectrum onto a fixed grid persists well beyond MS/MS [13], and how best to present a spectrum to a model remains open enough that serialising peaks as text for a language model is an active line of enquiry [12]. These systems outperform binned baselines, and the improvement is routinely read as evidence that the representation was the problem.

Binning has not gone unexamined. Giné et al. [7] benchmark featurization strategies for structure annotation across a large grid of models, comparing adaptive binning and spectrum hashing against learned embeddings, and report that retrieval degrades markedly as the mass tolerance is relaxed. Working on chemical-similarity prediction rather than structure annotation, de Jonge et al. [8] settle on 0.1 Da bins and report that finer bins did not improve performance. These are the two most direct empirical statements available about what mass precision is worth, and they do not agree with each other — which is itself a reason to ask the question in a form that admits a clean answer.

That reading does not follow from the experiments, and the reason is structural rather than careless. A binned vector and a peak set are different kinds of object: the first is dense and fixed-length, the second sparse and variable-length and permutation-invariant. No architecture reads both. Abandoning bins therefore forces a change of encoder, and a comparison of a peak-level model against a binned baseline varies the representation and the architecture together, usually alongside differences in training data and objective.

This applies to benchmarks as well as to individual systems. A study that compares many featurizations [7] measures each one bundled with the encoder that reads it — adaptive binning with a network that takes a dense vector, a learned embedding with the architecture that produced it — and so estimates the joint effect of representation and architecture at many points rather than separating them at any one. That is the right design for choosing a pipeline and the wrong one for attributing a gain. What is missing is not breadth but a contrast in which the mass axis moves and nothing else does.

This paper supplies the separation. The obstacle above is real but not insurmountable: while a peak-set encoder cannot read a binned vector, it can read peaks whose m/z has been *rounded to*

the bin grid. Quantising the mass axis — we use the term throughout for the precision of the m/z scale, the component of the input representation that binning actually destroys — while leaving the peak-set interface intact yields an arm that carries the baseline’s mass information through the treatment’s architecture, and inserting it between the two turns a confounded two-arm comparison into a decomposition. We report that decomposition across three training seeds, extend it with a bin-width sweep that prices precision as a continuous quantity, and examine what the results imply for how baselines in this area should be constructed and read.

Methods

Corpora and splits

Both corpora are drawn from the CASMI 2026 training corpus, an aggregation of eleven public and in-house MS/MS libraries whose `ingest_lib` column records each spectrum’s source.

Our primary corpus is the openly licensed subset: every spectrum contributed by GNPS, MassBank or MoNA, 414,992 spectra over 52,428 structures, spanning 66 instrument types and 112 adducts. We use it because it is the part of the corpus a third party can obtain from its sources without an account, and because it is acquisitionally diverse.

Our secondary corpus is a contiguous slice of two of the file’s twenty-one row groups, 262,144 spectra. Because the file is ordered by source library, that slice is not a sample of the corpus: it is 99.0% one in-house library and 1.0% another, and 99% of its spectra come from a single instrument (timsTOF), across two instrument types and sixteen adducts. We report it not as a second sample but as a deliberately contrasting one — a near-homogeneous corpus against a heterogeneous one — which is what makes the comparison in Fig. 2 informative.

Spectra were retained if the precursor m/z was below 2,000 and the reported precursor error within 20 ppm; peaks were retained above 0.1% of the base peak and below the precursor plus 1.5 Da, capped at the 256 most intense, with m/z held in double precision throughout preprocessing. On the open corpus this leaves 372,200 spectra over 50,287 structures and 24,627,256 peaks.

Structures were assigned to five folds by grouping generic Murcko scaffolds, with acyclic structures falling back to grouping by their own identifier, and fold sizes balanced greedily. Scaffold grouping is stricter than a random split and follows the generalisation-demanding practice argued for in MassSpecGym [9]; we did not adopt that benchmark’s own split because our corpora are selected by source library rather than drawn from it. Folds 2–4 were used for fitting, fold 1 for model selection, and fold 0 was held out; all reported numbers are on fold 0 (10,058 molecules on the open corpus, 8,692 on the single-library corpus).

Structural targets

Each structure was decomposed with BRICS and represented by the presence or absence of each of the 512 most frequent attachment-labelled fragments, the vocabulary fitted on the training folds alone. On the open corpus the target matrix is 99.3% zero with a median of 4 positive targets per molecule; 433 targets carry at least five held-out positives, at a mean prevalence of 0.0085, which is the chance level for macro AUPRC. The single-library corpus is less sparse (97.1% zero, median 17, 477 scored targets, chance 0.0093), reflecting how much more chemical variety three public libraries carry than one in-house library. Because almost all targets fall below 1% prevalence we report AUPRC as the primary metric and AUROC only as a secondary summary.

The five arms

The design is a ladder — M0, M1D, M1R, M1, M2 — in which each arm differs from the one below it in exactly one respect. We describe M1R before M1D because M1R is the control the design turns on, and M1D subdivides the effect M1R isolates.

M0, the conventional baseline. Each spectrum is projected onto 2,048 fragment bins and 2,048 precursor-minus-fragment bins of width 0.5 Da over $[0, 1024)$, with square-root intensity accumulated into each bin and the concatenated vector L_2 -normalised. The 4,096-dimensional vector is encoded by a two-layer perceptron. This is the representation used, with minor variation in bin width, throughout the MS/MS machine-learning literature.

M1R, the rounding control. Each peak’s m/z is assigned to a cell by $\lfloor m/w \rfloor$ with $w = 0.5$ Da — the same indexing M0’s encoder uses — and placed at that cell’s centre. Peaks sharing a cell are merged by accumulating square-root intensity, the accumulator M0 applies, so the merged peak carries the total M0 would have seen. The resulting tokens pass through the peak-set encoder described below. The arm therefore holds mass information at M0’s resolution while using M1’s architecture.

One detail is load-bearing. The encoder derives the neutral loss as $m^{\text{prec}} - m$. Quantising the peaks while leaving the precursor at full precision would return the precursor’s mass defect to every token through that subtraction, silently defeating the control, so the precursor is snapped to the same grid whenever quantisation is active. A unit test asserts that two precursors in one cell give identical embeddings under the control and different embeddings without it.

M1D, the interface control. Identical to M1R except that self-attention is removed from every block, leaving a position-wise network over the same peak tokens with the same acquisition conditioning, pooled by a masked mean rather than a learned token. Removing attention frees 66,048 weights per block, so the feed-forward is widened in this arm alone to 548 units; M1D therefore carries 714,528 parameters against M1R’s 712,336, above rather than below. The arm separates the change of interface from the attention mechanism: peaks presented individually rather than summed into a dense vector, with no token able to see any other.

M1, full precision. Identical to M1R with quantisation disabled, so peaks reach the encoder at the precision the instrument reported. M1R and M1 are identical in every other configuration field and have identical parameter counts (712,336); a test asserts both.

M2, hierarchical aggregation. Identical to M1 except that mean pooling over a molecule’s spectra is replaced by a measurement model weighting each spectrum by a learned, acquisition-dependent precision. This arm is included to give the other two effects a scale, not as a contribution of this paper.

Peak encoding

Each peak is encoded as a token. For a peak at m/z m in a spectrum with precursor m^{prec} we form a multi-scale sinusoidal embedding

$$\varphi_L(m) = [\sin(2\pi m/\lambda_1), \cos(2\pi m/\lambda_1), \dots, \sin(2\pi m/\lambda_L), \cos(2\pi m/\lambda_L)]^\top, \quad (1)$$

with $L = 32$ wavelengths spaced geometrically from 2 mDa to 10^3 Da, applied both to m and to the neutral loss. Short wavelengths resolve the mass defect and long wavelengths carry nominal mass. The token also receives the explicit defect $\delta = m - \text{round}(m)$, square-root and log-transformed intensity, and intensity rank.

At a wavelength of 2 mDa and m/z 342 the phase in Eq. (1) exceeds 10^6 , and evaluating it in

single precision destroys exactly the low-order information the embedding exists to preserve. We therefore compute the phase in double precision and reduce it modulo 2π before casting. With this reduction the embeddings of $\text{C}_4\text{H}_3\text{O}^+$ and C_5H_7^+ , which share a 0.5 Da bin, differ by 1.82 in the infinity norm.

Tokens are encoded by a four-block Transformer with a learned pooling token, each block conditioned on the acquisition covariates (adduct, polarity, instrument family, harmonised collision energy) by adaptive layer normalisation. The encoder is permutation invariant and accepts variable peak counts.

Bin-width sweep

To price precision as a continuous quantity rather than at one grid, we repeated the M1R construction at $w \in \{0.25, 0.1, 0.05, 0.01\}$ Da, with the encoder, parameter count, schedule and seed fixed. Together with $w = 0.5$ (M1R) and full precision (M1) this gives six points differing only in how finely the mass axis is resolved. The sweep was run on the single-library corpus only.

Training protocol

Every arm was trained for fifteen epochs with AdamW, a one-cycle schedule peaking at 10^{-3} , batches of 16 molecules, mixed precision and gradient clipping at 1.0. The loss was binary cross-entropy over the 512 presence targets with an auxiliary negative-binomial term on 88 functional-group counts, and checkpoints were selected on validation macro AUPRC. No hyperparameter search was performed for any arm. The ladder arms were each trained three times under seeds $\{2026, 7, 13\}$: five arms on the open corpus and four on the single-library corpus, twenty-seven training runs in all, plus the sweep arms once each.

Evaluation

We report macro AUPRC over targets with at least five held-out positives, with macro AUROC, micro AUPRC and Brier score as secondary summaries. Because arms are evaluated on identical molecules and identical targets, the appropriate contrast is paired: each target’s average precision is first averaged over the three seeds, the per-target differences are then taken, and their mean is bootstrapped over targets (5,000 resamples). We report the fraction of targets improved alongside, and, for each effect, its standard deviation across seeds — which, unlike the bootstrap interval, reflects initialisation variance. Where an ordering between effects is at issue we check it within each seed rather than reading it off the means.

Results

The five arms

The design is a ladder in which each arm differs from the one below it in exactly one respect (Table 1). M0 is the conventional baseline: a 0.5 Da binned intensity vector read by a two-layer perceptron, with the acquisition covariates concatenated to that vector so the baseline sees the same conditioning information as every other arm (Sect.). M1D keeps that binned mass axis but presents the spectrum as a set of peak tokens rather than one dense vector: each peak is snapped to its bin centre, colliding peaks are merged, and the tokens pass through a position-wise network with no attention, pooled by a masked mean. M1R adds self-attention to those same tokens. M1 hands the same attentive encoder the peaks at full precision. M2 adds hierarchical pooling over a molecule’s spectra. M1R and M1 are identical in every configuration field but the width of the mass

grid, and carry identical parameter counts; M1D differs from M1R only in carrying no attention, with its feed-forward widened so that it is not also the smaller arm (714,528 parameters against 712,336). Each arm was trained three times, under seeds {2026, 7, 13}.

M1R is the rung that makes the experiment mean anything. Without it, the contrast between a binned baseline and a peak-set model changes the mass axis and the encoder together, and a difference between them cannot be assigned to either. M1D then splits the encoder itself, separating the change of interface — peaks presented individually rather than summed into a dense vector — from the attention built on top of it.

Table 1: The five arms on the open corpus, each differing from the one above it in a single respect. Macro AUPRC on 10,058 held-out molecules, mean over three seeds. Chance is 0.0085 over 433 scored targets.

arm	mass axis	encoder	pooling	parameters	macro AUPRC	SD over seeds
M0	0.5 Da bins	dense vector, MLP	mean	1,172,752	0.1418	0.0013
M1D	0.5 Da bins	peak tokens, no attention	mean	714,528	0.1590	0.0024
M1R	0.5 Da bins	peak tokens, attention	mean	712,336	0.1567	0.0025
M1	full precision	peak tokens, attention	mean	712,336	0.1709	0.0022
M2	full precision	peak tokens, attention	hierarchical	829,584	0.1888	0.0015

Half of the apparent advantage is the baseline’s missing metadata

Our first version of this experiment reproduced the standard baseline faithfully, and that turned out to include a handicap we had not intended. The binned encoder took the acquisition covariates as an argument and never used them, so M0 alone could not see the collision energy, adduct, polarity or instrument, while every peak-set arm conditioned each of its blocks on them. Collision energy largely determines which bonds break, so a model blind to it must marginalise over every energy in the corpus.

Concatenating the same covariate embedding to the binned vector raises M0 from 0.1130 to 0.1418, an improvement of +0.0288 — larger than any of the three effects this paper set out to measure. The gap a two-arm comparison would report falls with it, from +0.0579 to +0.0291. Half of what looked like the advantage of reading peaks at full precision was the baseline not being told how the spectrum was acquired.

We report this because it is the paper’s own thesis turned on the paper. Every number below is measured against the conditioned baseline.

Against a fair baseline, the encoder and the mass axis are worth the same

Replacing the binned vector with a full-precision peak set raises macro AUPRC from 0.1418 to 0.1709 against a chance level of 0.0085, a paired per-target gain of +0.0291 (95% CI +0.0208 to +0.0371) improving 72.5% of targets. The control splits it (Fig. 1, Table 2). Changing the encoder alone, with the mass axis binned on both sides, is worth +0.0149 (+0.0080 to +0.0221; 66.7% of targets). Changing the mass axis alone, with the same encoder on both sides, is worth +0.0142 (+0.0097 to +0.0187; 64.9% of targets). The two sum to the joint effect exactly, as they must, since each target’s contribution enters both sides linearly.

The two halves are therefore indistinguishable: 51% encoder and 49% mass axis. Against the unconditioned baseline the same decomposition read 76% and 24%. Both components remain real, each interval excluding zero and each improving a clear majority of targets, but the claim that one dominates the other does not survive a baseline that has been told the collision energy.

Within the encoder, attention contributes nothing

The encoder effect itself divides further. Presenting the spectrum as peak tokens rather than one dense vector, with no attention anywhere in the network, is worth +0.0172 (+0.0102 to +0.0242; 58.0% of targets), and it holds in all three seeds. Adding self-attention to those same tokens is worth -0.0023 (-0.0066 to +0.0020), an interval straddling zero, improving 54.3% of targets — barely more than half — and positive in only one seed of three. The two sum to the encoder effect exactly.

The interface accounts for 115% of the encoder effect and attention for -15%. We do not read the negative sign as evidence that attention hurts; we read the interval as evidence that it does nothing measurable here. What matters is that a position-wise perceptron over peak tokens captures the entire advantage that a four-block Transformer over the same tokens captures, at a parameter count slightly in its favour.

This resolves a question the design left open in its first form. The contrast between M0 and M1R changed both the interface and the network family, so part of what we called “the encoder” might have been either. It is the interface.

Table 2: Paired per-target differences on the open corpus, bootstrapped over targets after averaging each target’s average precision across the three seeds. The interface and attention rows decompose the encoder row; the attention interval includes zero. The final column is the standard deviation of the effect across seeds.

contrast	isolates	Δ AUPRC	95% CI	targets improved	SD over seeds
M1D – M0	interface	+0.0172	[+0.0102, +0.0242]	58.0%	0.0016
M1R – M1D	attention	-0.0023	[-0.0066, +0.0020]	54.3%	0.0023
M1R – M0	encoder	+0.0149	[+0.0080, +0.0221]	66.7%	0.0013
M1 – M1R	mass axis	+0.0142	[+0.0097, +0.0187]	64.9%	0.0004
M2 – M1	aggregation	+0.0179	[+0.0136, +0.0223]	63.7%	0.0009
M1 – M0	encoder and mass axis	+0.0291	[+0.0208, +0.0371]	72.5%	–
M2 – M0	all three	+0.0470	[+0.0377, +0.0566]	73.9%	–

What the ordering supports, and what it no longer does

Aggregation is now the largest of the three effects, exceeding the encoder in all three seeds (+0.0009, +0.0030, +0.0049) and the mass axis in all three (+0.0028, +0.0045, +0.0037). It exceeds the encoder in all three seeds on the single-library corpus as well. That ranking we regard as established.

The encoder and the mass axis cannot be separated. Their difference is +0.0019, +0.0015 and -0.0012 across seeds: positive twice, negative once, and an order of magnitude smaller than either effect. It reverses on the single-library corpus too, in the opposite direction (-0.0016, +0.0006, -0.0090), so the failure to order them is not a property of one corpus. Against the unconditioned baseline the encoder led by +0.0296 and that ordering held in every seed. Conditioning the baseline erased the lead.

The mass-axis effect remains the most precisely measured quantity in the study, with a seed standard deviation of 0.0004 against 0.0013 for the encoder and 0.0009 for aggregation. That the smallest effects here are also the most reproducible is worth stating plainly: they are small effects measured precisely, not noise.

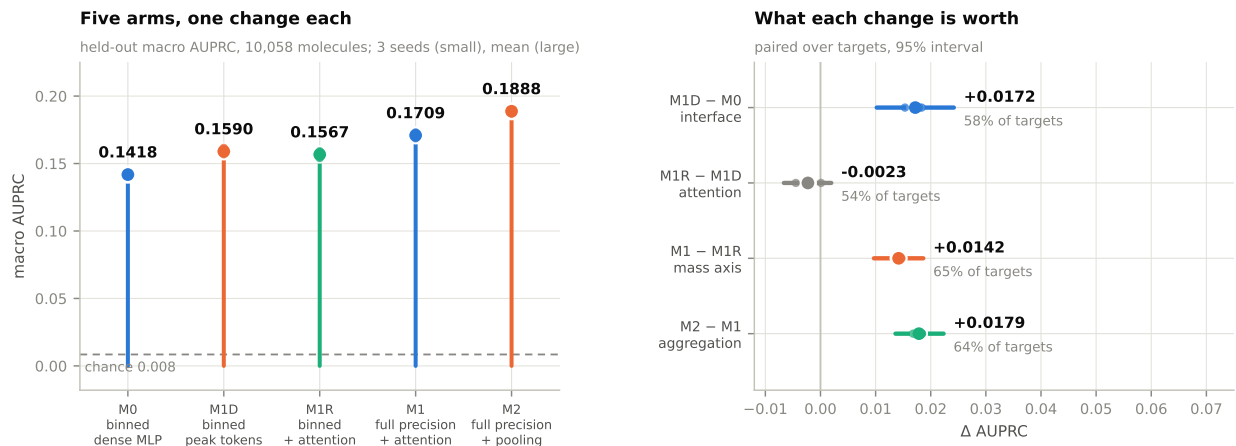


Figure 1: Decomposing the gain on the open corpus. **Left:** held-out macro AUPRC for the five arms; small dots are individual seeds, large dots their mean. **Right:** what each single change is worth, paired over targets with 95% bootstrap intervals; small dots are the per-seed effect sizes.

The same decomposition on a very different corpus

The finding is a claim about how the field should construct comparisons, not about one dataset. We therefore repeated the entire ladder on a second corpus with almost nothing in common with the first. That corpus is a single in-house library, 99% of whose spectra come from one instrument (timsTOF), against the open corpus’s 66 instrument types drawn from three public libraries (Fig. 2, Table 3).

Every effect is smaller on the open corpus, and by a similar factor: the encoder falls from +0.0234 to +0.0149, the mass axis from +0.0268 to +0.0142, and aggregation from +0.0322 to +0.0179. Both ladders were fitted against the conditioned baseline, so all three contrasts are comparable.

What replicates is the finding, not merely its direction. On the single-library corpus the encoder is worth +0.0234 and the mass axis +0.0268; their difference is -0.0016 , $+0.0006$ and -0.0090 across seeds, reversing sign exactly as it does on the open corpus. The encoder’s share of the gain a two-arm comparison would report is 46.7% here against 51.3% there. Two corpora with almost nothing in common — one library against three, two instrument types against 66, 16 adducts against 112 — return the same answer: the encoder and the mass axis are worth about the same, and neither can be ordered against the other. Aggregation exceeds the encoder in all three seeds on both.

The correction itself replicates too, and in a way that supports the reading we give it. Conditioning M0 on the acquisition covariates was worth +0.0288 on the open corpus and +0.0142 on the single-library one. If those covariates matter because they tell the model how the spectrum was produced, their value should scale with how much the production varied, and the open corpus carries 66 instrument types and 112 adducts against two and 16. We offer this as a consistency check rather than a result.

What binning removes, and how much of it matters

The mechanism is visible before any model is fitted. Of the 3,797,166 peaks in the open corpus’s held-out spectra, 23.6% fall into a 0.5 Da bin already occupied by another peak from the same spectrum and are therefore absorbed. On the single-library corpus’s held-out spectra the same figure is 34.4% (Fig. 3). The loss falls with the grid on both, reaching 2.9% and 6.5% respectively at 0.01 Da.



Figure 2: The decomposition on two corpora. **Left:** each effect measured on both, with per-seed values (small) and their mean (large). **Right:** the encoder’s share of the gain a two-arm comparison would attribute to the representation.

Table 3: The decomposition on two corpora. The single-library corpus is 99% timsTOF from one in-house library; the open corpus spans 66 instrument types across GNPS, MassBank and MoNA. Absorption is measured on each corpus’s own held-out spectra.

effect	single library	SD (single)	open corpus	SD (open)
encoder	+0.0234	0.0002	+0.0149	0.0013
mass axis	+0.0268	0.0049	+0.0142	0.0004
aggregation	+0.0322	0.0035	+0.0179	0.0009
encoder share of M1 - M0	46.7%	—	51.3%	—
peaks absorbed at 0.5 Da	34.4%	—	23.6%	—

What a merged peak costs is compositional identity. Enumerating plausible CHNOS compositions within 10 ppm places 328 candidate fragment formulas inside a single 0.5 Da bin at m/z 300. The width of the mass window governs that count, in the way Kind and Fiehn established for precursor formulas [10]. A bin three orders of magnitude wider than the instrument’s error therefore discards almost all of the discrimination the measurement bought.

The two corpora move in the same direction on both quantities. The one that loses more peaks to binning is also the one where the mass axis is worth more: 34.4% and +0.0268 against 23.6% and +0.0142. The correspondence is not proportional, since absorption falls to 0.69 of its value while the effect falls to 0.53. Peak absorption is therefore not a complete account of the cost, but it does move with it.

We offer one hypothesis for the difference, which our design does not test. The single-library corpus is almost entirely timsTOF, a high-resolution instrument on which every reported m/z carries real precision. The open corpus spans 66 instrument types of varying resolution, and on the lower-resolution ones there is less precision present to preserve, so the average value of preserving it falls. Stratifying the mass-axis effect by instrument resolution would test this directly and we regard it as the most informative experiment this work leaves undone.

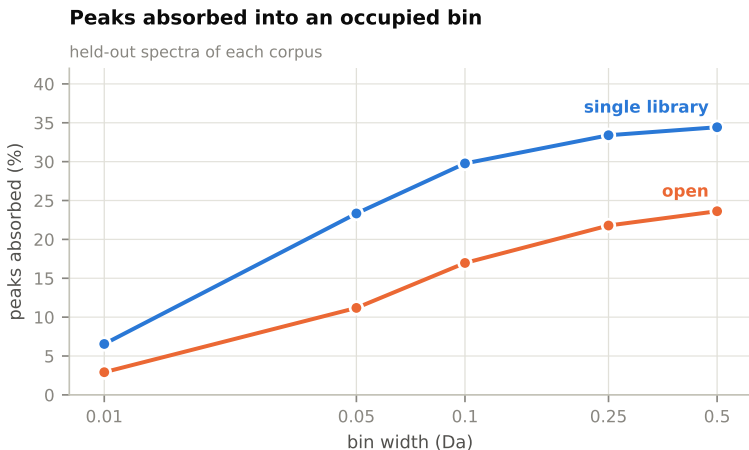


Figure 3: Fraction of peaks absorbed into an already-occupied bin, against grid width, on each corpus’s own held-out spectra.

How much precision is needed

The rounding control prices one particular grid. Sweeping the grid width on the single-library corpus, with the encoder and parameter count fixed throughout, prices precision as a function of how much of it is retained (Table 4). At 0.5, 0.25 and 0.1 Da the model sits between 0.0801 and 0.0867, a range of 0.0066 — under twice that arm’s seed standard deviation of 0.0039, and unordered within itself; coarsening from 0.1 to 0.5 Da costs nothing measurable. Below that the curve moves: 0.05 Da reaches 0.0909 and 0.01 Da reaches 0.1030, closing 65% of the distance to full precision. Even a 10 mDa grid — fine enough to separate the isobaric fragments that motivate the mass-defect argument — leaves roughly a third of the benefit unrealised, which suggests the model uses the defect as a continuous quantity rather than merely as a discriminator between candidate formulas.

Two cautions. Each width was run once, so the non-monotonic dip at 0.25 and 0.1 Da, spanning

0.0066, is unremarkable across five single runs and we read it as noise rather than structure. We also tested an alternative explanation. The sinusoidal basis has wavelengths that are integer divisors of these grid spacings, which would render some channels constant. But only one to three of thirty-two channels are affected at any width, and the count does not track performance ($r = -0.26$), so we discount aliasing. The sweep was run on the single-library corpus only; whether the 0.05 Da threshold moves with instrument diversity is open.

Table 4: Bin-width sweep, single seed. Every arm uses the M1 encoder at an identical parameter count; only the width of the mass grid changes. The final column expresses each result as a share of the distance between the 0.5 Da arm and full precision.

bin width (Da)	peaks absorbed	macro AUPRC	share of the 0.5 Da gap closed
0.5	37.5%	0.0867	0%
0.25	36.8%	0.0818	-19%
0.1	33.1%	0.0801	-26%
0.05	25.9%	0.0909	17%
0.01	7.5%	0.1030	65%
full precision	0.0%	0.1119	100%

Discussion

The headline is a reattribution, and it has three layers. A two-arm comparison on the open corpus credits the spectral representation with +0.0579 macro AUPRC. Half of that is the baseline not having been told how the spectrum was acquired. Of the +0.0291 that remains, the encoder takes +0.0149 and the mass axis +0.0142, a difference that reverses across seeds. And the encoder’s own share is entirely the switch to per-peak tokens: attention over those tokens is worth -0.0023 , with an interval straddling zero.

The order of magnitude is worth dwelling on. The largest single quantity we measured was not an architectural choice or a representational one. It was a missing input to the comparator, worth +0.0288 — more than the encoder, the mass axis and aggregation individually. We did not plant it; we reproduced a standard baseline faithfully and inherited it, and it survived four rounds of revision by authors actively looking for confounds in exactly this experiment.

It is worth being clear about why this went unnoticed, because the answer is not inattention. The confound is forced by the data structures involved. A binned vector is dense and fixed-length; a peak set is sparse, variable-length and permutation-invariant. No architecture reads both, so an experiment that changes only the representation appears impossible, and the comparison everyone can run is the one that changes both. The rounding control works not by finding a network that reads both objects but by degrading one object into the other while keeping the interface fixed. The manoeuvre is simple in retrospect, and we suspect it generalises: wherever a representation change forces an architecture change, asking whether the richer representation can be impoverished in place is likely to be more productive than looking for a universal encoder.

Our result does not diminish the systems that abandoned binning. CSI:FingerID, MIST, DreaMS and TransExION are better than binned baselines, and this work does not dispute it. What it changes is the account of *why*. Those systems differ from a binned baseline in both respects. To the extent that they do, our decomposition suggests the larger part of their advantage

comes from the architectures they adopted rather than from the mass information they preserved. More narrowly still, it suggests the gain lies in letting the model see peaks individually, not in any particular mechanism for relating them. That is a testable claim about each of them, and both controls we describe are cheap enough to apply.

The second corpus replicates all of this and adds a caveat of its own. The encoder and the mass axis are indistinguishable there too, and their difference reverses across seeds in the opposite direction, so the failure to order them belongs to the comparison rather than to one dataset. But every effect is smaller on the open corpus, the mass axis roughly halving from +0.0268 to +0.0142 as the instrument mix widens from one to 66. Our hypothesis for the mass-axis change is that a heterogeneous corpus contains many spectra whose reported m/z carries little real precision, so preserving precision helps less on average. If that is right, the value of full-precision input depends on the instrument mix a method will encounter, and a method tuned on a homogeneous high-resolution corpus will disappoint on a heterogeneous one for reasons that have nothing to do with its architecture. We did not test this, and say so in the Limitations; we flag it because it bears directly on how these methods should be benchmarked.

The sweep adds a second complication. Precision buys nothing measurable between 0.5 and 0.1 Da; the gain appears below 0.05 Da and is not exhausted at 0.01 Da. A 10 mDa grid separates the isobaric fragments that motivate the mass-defect argument, yet still leaves about a third of the benefit unrealised, which suggests the model uses the defect as a continuous quantity rather than merely as a discriminator between candidate formulas. This sits awkwardly with the common design choice, TransExION’s included, of embedding the defect alongside a coarsely quantised nominal mass: on our evidence the resolution of that channel matters more than its presence.

It also disagrees with a recent result we should not paper over. de Jonge et al. [8] adopt 0.1 Da bins for MS2DeepScore 2.0 and report that finer bins did not improve performance; our sweep finds nothing measurable between 0.5 and 0.1 Da, agreeing with them over that range, but a gain of +0.0121 between 0.05 and 0.01 Da, disagreeing below it. Three explanations seem live. The tasks differ — chemical-similarity prediction between two spectra may be satisfied by coarser features than inferring which substructures a molecule contains, since the former needs only that two spectra of similar molecules land near each other. The architectures differ. And our sweep is single-seed, so the disagreement may be ours. We cannot adjudicate this from our data, and flag it because a reader comparing the two papers will notice, and because the question of what bin width suffices is evidently not settled by either.

Limitations

The open corpus is roughly a sixth of the available spectra and the single-library corpus a tenth; absolute values will move with scale even if the relative sizes do not. Three seeds per arm are enough to establish that aggregation exceeds the encoder, which holds in every seed on both corpora, but not to order the encoder against the mass axis, whose difference changes sign across seeds on both. The mass axis and aggregation separate on the open corpus and not on the single-library one, and we declined to rank them there. The sweep carries one run per width and was run on one corpus, so only its endpoints should be read as findings and its 0.05 Da threshold may not transfer. The instrument-diversity hypothesis above is untested: stratifying the mass-axis effect by instrument resolution would settle it and is the most informative experiment this work leaves undone. Finally, the null result on attention is a null result, and we report it as one: the interval on -0.0023 straddles zero at this sample size and on this task, which is not the same as showing that attention cannot

help. A larger corpus, a different target family, or a task where relating peaks to one another matters more than identifying them individually could all move it.

The practical consequence concerns controls. A method evaluated against a binned baseline is compared against an arm that differs from it in at least two ways, and on this evidence those two are worth about the same — while a third difference, belonging to neither the representation nor the architecture, was worth more than either. A paper reporting a gain over such a baseline has not shown that its representation is responsible, whatever its discussion says. The remedy is inexpensive: report the comparator’s encoder as explicitly as its representation, and where the claim is about representation, include an arm that changes only that.

Conclusion

Binned intensity vectors entered MS/MS machine learning from spectral library search, and the field’s move away from them has been read as evidence that the spectral representation was the limiting factor. That reading was never tested, because no architecture reads both a binned vector and a peak set, so representation and encoder have always changed together. Inserting an arm that carries binned mass information through a peak-set encoder separates them. On an openly licensed corpus of 414,992 spectra spanning 66 instrument types, across three seeds and 10,058 held-out molecules, a binned baseline trails a peak-set model by +0.0579 macro AUPRC. Half of that closes once the baseline is given the acquisition covariates the peak-set arms had all along. The remaining +0.0291 divides evenly between the encoder (+0.0149) and the mass axis (+0.0142), by a margin that reverses across seeds. Within the encoder, per-peak tokens account for the whole effect and attention for none (−0.0023, interval straddling zero), and hierarchical aggregation (+0.0179) exceeds both. Mass precision matters, and matters more the more homogeneous and higher-resolution the instruments are. But what the binned baseline most conspicuously lacked here was not bins, an encoder, or precision. It was an input nobody had noticed withholding.

The control that shows this is cheap and not specific to mass spectrometry. Wherever a change of representation forces a change of architecture — and that is the common case, not a rare one — the richer representation can usually be impoverished in place while the interface is held fixed, turning a confounded comparison into a decomposition. We would encourage others to try it on their own baselines.

Supplementary Information

Additional file 1 (PDF) reports per-arm and per-seed metrics for all twenty-seven training runs, the per-seed effect sizes and the within-seed ordering checks behind every ranking claimed here, the full bootstrap distributions, the peak-absorption analysis broken down by adduct, collision energy and instrument family, the bin-width sweep in full, and the exact commands that regenerate all of it. Per-target average precision for every arm and seed, the bootstrap draws and the stratified absorption counts are provided as machine-readable CSVs in the repository under **results/**.

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Author contributions

X.P. designed the study, implemented the models and the control, performed the experiments and analyses, and wrote the manuscript. C.W. contributed to the interpretation of the chemical results and revised the manuscript. Both authors read and approved the final manuscript.

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Availability of data and materials

All code, configurations, trained-run reports and the analysis that generates every number in this paper are available at <https://github.com/xulinpan/cheminformatics> under the MIT licence, including the script that builds the open subset and a document recording each source library’s contribution. The primary corpus consists of the spectra contributed by GNPS, MassBank and MoNA. All three are openly licensed and obtainable from those sources. We selected them from the aggregated CASMI 2026 file, which the competition organisers distribute; the in-house libraries that file also contains are not redistributable, and are not used in our primary experiments.

Declarations

Competing interests

The authors declare no competing interests.

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